

PYTHON QUESTIONS OF CONSTRUCTOR AND SELF :-

1. Can we define multiple `__init__()` methods in the same class?
2. Is `self` a keyword in Python?
3. What is MRO in Python? How can you check the MRO of a class?
4. Why is `self` required as the first parameter of an instance method?
5. What is the difference between a default constructor and a parameterized constructor?
6. What happens if `__new__()` does not return an object?
7. What is the difference between `self` and a normal method parameter?
8. What is constructor chaining?
9. Can `self` be used inside a `@staticmethod`?
10. Can `__new__()` return an object of a different class ?
11. What is the purpose of `super().__init__()`?
12. What is the difference between:

`super().__init__()`

and

`Parent.__init__(self)`

13. Why is the Method Resolution Order (MRO) required, and how is it related to constructors?
14. If two parent classes have the same method, which method will Python execute?
15. Does Python support constructor overloading? If not, how can we achieve similar behavior?
16. How does MRO affect constructor calls when multiple classes have `__init__()`?
17. Can we use another name instead of `self`?

18. What happens if you accidentally modify self inside a method?
19. Why does Python explicitly require self, while languages like Java and C++ don't explicitly write it in every instance method?
20. Why does __new__() receive cls, while __init__() receives self?
21. Can self be used inside a @classmethod?
22. How does Python maintain a consistent order in the case of the Diamond Problem?
23. Why does a class method use cls instead of self?
24. In what order are __new__() and __init__() executed when an object is created?
25. What is the difference between __new__() and __init__()?
26. Why is __new__() particularly important when working with immutable objects?
27. Can __init__() return a value? What happens if it returns something other than None?
28. Does super() always call the immediate parent class?
29. Can we pass one object as self to another object's method?
30. What is the difference between self.age and age inside a method?